

TUSHAR VAIBHAV

Backend Engineer | .NET Core and Node.js | Microservices | SQL/NoSQL | REST APIs | Full-Stack Developer

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tusharvaibhavmca2020.github.io/Portfolio

PROFESSIONAL SUMMARY

Backend Engineer with 4 years of experience designing scalable, high-performance systems using C#, ASP.NET Core, Node.js, REST APIs, SQL Server, PostgreSQL, and MongoDB. Proven strength in microservices architecture, enterprise data migration, ETL pipelines, database optimization, and full-stack delivery with ReactJS and Angular. Experienced across airline, carbon-analytics, forensic, and agri-distribution domains. Skilled at using AI tools such as Claude and ChatGPT to accelerate development, debugging, and solution design.

TECHNICAL SKILLS

Languages: C#, JavaScript, TypeScript, SQL, Python

Backend: ASP.NET Core, Web API, MVC, Razor Pages, Entity Framework Core, Node.js, Express.js

Frontend: ReactJS, Angular, RxJS, HTML, CSS

Databases: SQL Server, PostgreSQL, MongoDB, Redis; data modeling, query optimization, indexing, stored functions (PL/pgSQL), aggregation pipelines

Architecture: Microservices, REST APIs, API Gateway (YARP), Event-Driven (MassTransit, RabbitMQ), SignalR, Layered Architecture, Dependency Injection, SOLID

Data and Integration: ETL/ELT, Data Migration, Schema Transformation, Batch and Bulk Processing, Salesforce Bulk API, Heroku Connect

Cloud and DevOps: Microsoft Azure, Docker, AWS S3, Firebase, JWT, 2FA, Git, TFS, BitBucket, Postman, IIS

AI-Assisted Development: Claude, ChatGPT, Ollama (local LLM), Prompt Engineering

WORK EXPERIENCE

Application Engineer, Quadrafort Technologies Pvt. Ltd., Noida, UP

September 2024 to Present

III Field Sales and Distribution Platform (Client: Insecticides India Ltd.) - Backend Developer (Node.js, PostgreSQL)

- Refactored the data-access layer of a Node.js and Express field-sales platform running on PostgreSQL by converting 565 or more inline SQL queries across 38 modules into parameterized PostgreSQL (PL/pgSQL) stored functions.
- Reduced complex hand-built queries (including 180-line multi-tier UNION queries) to single parameterized function calls, moving query logic to the database layer to improve maintainability, reuse, and performance.
- Kept all queries fully parameterized to guard against SQL injection and preserved existing API behavior with no regressions throughout the migration.

Enterprise Data Migration Platform (Client: LTIMindtree) - Backend / API Developer (ASP.NET Core)

- Designed high-performance ASP.NET Core REST APIs to migrate [ADD number] million records across [ADD number] tables from SQL Server to MongoDB with zero data loss.
- Converted complex SQL joins into optimized MongoDB aggregation pipelines, improving query efficiency by [ADD percentage].
- Implemented end-to-end ETL covering schema transformation, ObjectId to legacy-ID mapping, and data-type conversions; built bulk-insert and batch-processing strategies for reliable high-volume processing.
- Resolved critical production issues including high CPU utilization and aggregation timeouts by refactoring merge stages and temporary-collection usage.

CarbonCare - Full-Stack Developer (ASP.NET Core, ReactJS)

- Built a secure ASP.NET Core and Razor Pages backend with REST APIs for a carbon-emission tracking platform.

- Created background scheduled services to process Excel uploads and auto-calculate emission metrics; developed responsive ReactJS dashboards with data visualizations; deployed and maintained the application on IIS.

eForensics FSL (Client: NIC) - Backend Developer (ASP.NET MVC)

- Integrated MongoDB into an existing PostgreSQL-based ASP.NET MVC application for hybrid storage; developed and optimized aggregation pipelines while maintaining data consistency with legacy systems.

Associate Application Engineer (.NET), Sutraa Technosoft Pvt. Ltd., Noida, UP

May 2024 to August 2024

- Built backend architecture with ASP.NET Core and Razor Pages, and developed dynamic ReactJS frontend components with integrated data-visualization dashboards.

Associate Software Engineer, Birdres Technologies Pvt. Ltd., Gurgaon, HR

June 2022 to May 2024

Eagle - Airline Information Platform - Backend Developer (ASP.NET Core Web API)

- Developed and maintained scalable ASP.NET Core Web APIs delivering real-time airline schedules, pricing, routes, and seat availability to [ADD number] concurrent users.
- Integrated multiple third-party airline systems with robust error handling, logging, and retry mechanisms.

CUSS - Common Use Self Service - Full-Stack Developer (AngularJS, SQL Server)

- Integrated multiple airline APIs to automate boarding-pass and bag-tag printing at airport self-service kiosks.
- Built real-time monitoring and reporting tools with AngularJS and SQL Server, improving operational transparency for airport operations teams.

KEY ENGINEERING PROJECTS

API Performance Analyzer - Multi-Tenant API Monitoring Platform (ASP.NET Core 8, Angular, MongoDB)

- Architected a 4-microservice ASP.NET Core 8 system behind a YARP API gateway with per-service MongoDB databases; a self-hosted APM platform that ingests HTTP performance logs and surfaces latency, error-rate, and traffic analytics.
- Built real-time dashboards with SignalR streaming and an Angular SPA; implemented batch ingestion (up to 1,000 entries per call), per-key rate limiting (500 requests per minute), and an in-memory metrics-aggregation engine computing latency percentiles, requests-per-second, and uptime.
- Added JWT and TOTP two-factor authentication, role-based access control, a configurable alerting engine (email and signed webhooks), anomaly detection, SLA tracking, and public status pages.

Enterprise Data Migration Engine - SQL/File to Salesforce (.NET 8 Microservices)

- Built a .NET 8 microservices platform (YARP gateway and MassTransit/RabbitMQ event bus) migrating data from SQL Server, PostgreSQL, MySQL, CSV, and JSON into Salesforce using Bulk API 2.0.
- Engineered a streaming, bounded-memory ETL pipeline (10,000-row batches via IAsyncEnumerable) with a pluggable mapping engine, foreign-key relationship reconstruction using Salesforce External Ids, per-row failure capture with retry-failed-only, cron scheduling, and throughput tracking.
- Integrated a local LLM (Ollama) for AI-assisted column-to-field auto-mapping; Angular control plane with PostgreSQL 16 and Docker.

EDUCATION

- Master of Computer Applications (MCA), School of Management Sciences, Varanasi, 2019 to 2022, 84%
- Bachelor of Computer Applications (BCA), L N College Bhagwanpur, Vaishali, Bihar, 2015 to 2018, 78.25%

CERTIFICATIONS

- MongoDB Query Optimization Techniques, MongoDB
- Problem Solving (Basic), HackerRank